Data reduction

Data reduction is the transformation of numerical or alphabetical digital information derived empirically or experimentally into a corrected, ordered, and simplified form. The purpose of data reduction can be two-fold: reduce the number of data records by eliminating invalid data or produce summary data and statistics at different aggregation levels for various applications.

The data reduction approach decreases the amount of data while preserving its integrity. Data reduction has no effect on the outcome of data mining, which implies that the outcome of data mining before and after data reduction is the same (or almost the same)

Why Data Reduction?

Data reduction process reduces the size of data and makes it suitable and feasible for analysis. In the reduction process, integrity of the data must be preserved and data volume is reduced. There are many techniques that can be used for data reduction.

Difference between Data Redundancy and Data Inconsistency

Data Redundancy :

It is defined as the redundancy means duplicate data and it is also stated that the same parts of data exist in multiple locations into the database. This condition is known as Data Redundancy.

Problems with Data Redundancy :

Here, we will discuss the few problems with data redundancy as follows.

1. Wasted Storage Space.
2. More Difficult Database Update.
3. It will lead to Data Inconsistency.
4. Retrieval of data is slow and inefficient.

Example – Let us take an example of a cricket player table.

Step-1 :

Consider cricket player table as follows.

Player Name	Player Age	Team Name	Team ID
Virat Kohli	32	India	1
Rohit Sharma	34	India	1
Ross Taylor	37	New Zealand	2
Shikhar Dhawan	35	India	1
Kane Williamson	30	New Zealand	2

Step-2 :

We can clearly see that the Team Name and Team Id are repeated at multiple places. we can make a separate table to store this information and reduce data redundancy.

Player Name	Player Age	Team Id
Virat Kohli	32	1
Rohit Sharma	34	1
Ross Taylor	37	2
Shikhar Dhawan	35	1
Kane Williamson	30	2

Step-3 :

This is known as Normalization used to reduce Data Redundancy.

Team Id	Team Name
1	India
2	New Zealand

Data Inconsistency:

When the same data exists in different formats in multiple tables. This condition is known as Data Inconsistency. It means that different files contain different information about a particular object or person. This can cause unreliable and meaningless information. Data Redundancy leads to Data Inconsistency.

Example –

If we have an address of someone in many tables and when we change it in only one table and in another table it may not be updated so there is the problem of data inconsistency may occur.

Differences:

Topic	Data Redundancy	Data Inconsistency
Condition	It will be applicable when the duplicate data exists in multiple places in the database.	It will be applicable when the duplicate data exists in different formats in multiple tables.
How to minimize it?	we can use normalization to minimize Data Redundancy.	we can use constraints on the database to minimize Data Inconsistency

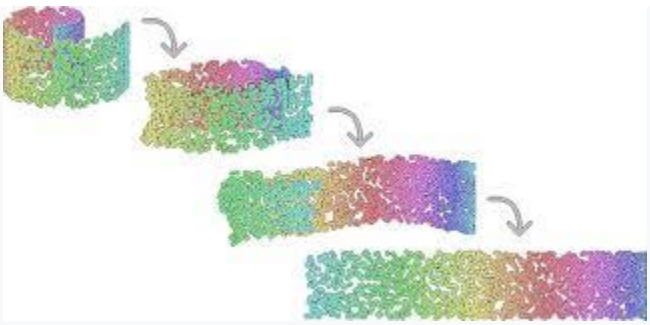
Objectives of Data Reduction

1. **Increases Building Efficiency:** Data analytics helps to reduce construction time and material-related costs by providing clear data and identifying potential structural errors before they happen.
2. **Reduces Environmental Impact:** The construction industry is one of the biggest polluters in the world. So we need to use eco-friendly building materials and do sustainable construction practices. Data analysis can help to solve this problem.
3. **Promotes Collaboration:** Lack of communication is one of the biggest problems construction professionals face while doing their duties. Fortunately, data analysis makes information accessible and easily shareable between crew members.
4. **Improves Working Conditions:** Data collected from health tracking software is used to improve working conditions on site.

Methods of Data Reduction

Dimensionality Reduction

When dimensionality increases, data becomes increasingly sparse while density and distance between points, critical to clustering and outlier analysis, becomes less meaningful. Dimensionality reduction helps reduce noise in the data and allows for easier visualization, such as the example below where 3-dimensional data is transformed into 2 dimensions to show hidden parts. One method of dimensionality reduction is wavelet transform, in which data is transformed to preserver relative distance between objects at different levels of resolution, and is often used for image compression.



An example of dimensionality reduction.

Numerosity Reduction

Numerosity Reduction is a data reduction technique which replaces the original data by smaller form of data representation. There are two techniques for numerosity reduction- **Parametric** and **Non-Parametric** methods.

This method of data reduction reduces the data volume by choosing alternate, smaller forms of data representation. Numerosity reduction can be split into 2 groups: parametric and non-parametric methods. Parametric methods (regression, for example) assume the data fits some model, estimate model parameters, store only the parameters, and discard the data. One example of this is in the image below, where the volume of data to be processed is reduced based on more specific criteria. Another example would be a log-linear model, obtaining a value at a point in m-D space as the product on appropriate marginal subspaces. Non-parametric methods do not assume models, some examples being histograms, clustering, sampling, etc.

